Unit Root and Stationarity Analysis

ADF and KPSS Tests -- O2 Step 2 Report

Intelligent National Housing Cost Estimating System (iNHCES)

TETFund NRF 2025 | Department of Quantity Surveying, ABU Zaria

Document:	O2_04_Stationarity_Analysis.pdf
Objective:	O2 -- Macroeconomic Variable Analysis (TETFund NRF 2025)
Variables:	7 macro series -- GDP, CPI, lending rate, Brent crude, NGN/USD/EUR/GBP
Tests:	ADF (Said & Dickey, 1984) + KPSS (Kwiatkowski et al., 1992)
Sample:	Annual panel 2000-2024 (up to 25 observations per series)
Integration:	I(0): 0 vars | I(1): 4 vars | Ambiguous: 3 vars
Implication:	VECM recommended
Results CSV:	02_macro_analysis/data/processed/stationarity_results.csv
Date:	25 April 2026

DATA SOURCE DECLARATION -- READ BEFORE USE

DATA SOURCE: PARTIALLY SYNTHETIC -- World Bank is LIVE; EIA Brent and CBN FX are SYNTHETIC unless API keys were set

INPUT DATA SOURCES FOR THIS ANALYSIS:

* GDP Growth, CPI Inflation, Lending Rate -- World Bank Data API (LIVE). Test statistics for these 3 variables are based on real official data.
* Brent Crude Oil -- SYNTHETIC fallback (no EIA_API_KEY or FRED_API_KEY). Hardcoded annual estimates from published EIA/BP statistical reviews.
* NGN/USD, NGN/EUR, NGN/GBP -- SYNTHETIC fallback (no FRED_API_KEY). Hardcoded annual estimates from CBN Annual Reports and IMF IFS database.

IMPACT ON RESULTS:

* ADF and KPSS statistics for Brent crude and FX series are derived from synthetic values. The DIRECTION (I(1) for exchange rates and oil prices) is consistent with macroeconomic literature for emerging economies, but exact statistics should be treated as illustrative until live data is available.

TO UPGRADE TO FULLY LIVE DATA:

1. Set EIA_API_KEY (eia.gov/opendata) and FRED_API_KEY (fred.stlouisfed.org)
2. Re-run: fetch_eia_oil.py and fetch_cbn_fx.py
3. Re-run: stationarity_analysis.py

REFERENCES:

Said & Dickey (1984). Biometrika, 71(3), 599-607. [ADF test]
Kwiatkowski et al. (1992). Journal of Econometrics, 54(1-3), 159-178. [KPSS]
Maddala & Kim (1998). Unit Roots, Cointegration, and Structural Change. CUP.

1. Unit Root Testing Methodology

Stationarity testing is a prerequisite for multivariate time series modelling. A non-stationary series (one with a unit root) included in a VAR model in levels produces spurious regressions with inflated R-squared values and invalid t-statistics. Correct identification of the integration order determines whether to use a VAR in differences, a VECM, or a combination of both.

1.1 Augmented Dickey-Fuller (ADF) Test

H0: The series has a unit root (non-stationary). H1: Stationary.

Lag length chosen by AIC (max 4 or (n-1)/3). Constant included (no trend).

DECISION: Reject H0 if p-value < 0.05 -- series is stationary at this transformation.

1.2 KPSS Test (Kwiatkowski-Phillips-Schmidt-Shin, 1992)

H0: The series is level-stationary. H1: Unit root present.

Reverses the null, reducing Type II error risk. Bandwidth set automatically.

DECISION: Reject H0 if p-value < 0.05 -- series has unit root.

NOTE: statsmodels bounds the KPSS p-value at [0.01, 0.10]; displayed values at boundaries should be read as '<= 0.01' or '>= 0.10'.

1.3 Integration Order Decision Rules

ADF	KPSS	Order	Interpretation
Reject (p<0.05)	Don't reject	I(0)	Stationary -- use in levels
Don't reject	Reject (p<0.05)	I(1)	Unit root -- difference or VECM
Reject (p<0.05)	Reject (p<0.05)	Contradiction	Structural break likely
Don't reject	Don't reject	Uncertain	Treated as I(1) conservatively

SMALL SAMPLE NOTE: With n = 25 annual observations, ADF and KPSS tests have limited power. Elliott, Rothenberg & Stock (1996) recommend ERS point-optimal tests for small samples. Results here are indicative. Final published paper should apply Zivot-Andrews (1992) break-adjusted tests and possibly Ng-Perron (2001).

2. ADF Test Results -- Levels and First Differences

ADF: H0 = unit root. Reject H0 (p < 0.05) = stationary. 5%CV = ADF 5% critical value (reject H0 if test stat < 5%CV).

Variable	Lev Stat	Lev p	5%CV(L)	ADF Lev Dec.	D1 Stat	D1 p	5%CV(D1)	ADF D1 De
GDP Growth Rate (Annu	-2.5519	0.1033	-2.992	unit root	-5.8719	0.0	-3.005	STATIONARY
CPI Inflation Rate (A	-0.8965	0.7891	-2.992	unit root	-5.194	0.0	-2.998	STATIONARY
Lending Interest Rate	-1.437	0.5644	-2.998	unit root	-4.7178	0.0001	-3.005	STATIONARY
Brent Crude Oil (USD/	-2.0549	0.263	-2.992	unit root	-4.3043	0.0004	-3.005	STATIONARY
NGN/USD Exchange Rate	1.6473	0.998	-3.022	unit root	3.1119	1.0	-3.013	unit root
NGN/EUR Exchange Rate	1.4755	0.9974	-3.013	unit root	2.6577	0.9991	-3.013	unit root
NGN/GBP Exchange Rate	1.8047	0.9984	-3.022	unit root	3.4476	1.0	-3.013	unit root

STATIONARY = ADF rejects H0 at 5% significance level. (L) = at Levels. (D1) = at First Differences. 5%CV = ADF 5% critical value (MacKinnon, 1994). Test statistics below the critical value support stationarity.

3. KPSS Test Results -- Levels and First Differences

KPSS: H0 = stationary. Reject H0 (p < 0.05) = unit root. Final column shows integration order from ADF + KPSS consensus.

Variable	Lev Stat	Lev p	KPSS Lev Dec.	D1 Stat	D1 p	KPSS D1 Dec.	Order
GDP Growth Rate (Annu	0.5245	0.0361	UNIT ROOT	0.3011	0.1	stationary	I(1)
CPI Inflation Rate (A	0.3893	0.0818	stationary	0.2399	0.1	stationary	I(1)
Lending Interest Rate	0.5625	0.0276	UNIT ROOT	0.0885	0.1	stationary	I(1)
Brent Crude Oil (USD/	0.271	0.1	stationary	0.1556	0.1	stationary	I(1)
NGN/USD Exchange Rate	0.6564	0.0175	UNIT ROOT	0.4723	0.0479	UNIT ROOT	I(2)*
NGN/EUR Exchange Rate	0.6831	0.0151	UNIT ROOT	0.4365	0.0614	stationary	I(2)*
NGN/GBP Exchange Rate	0.6388	0.0191	UNIT ROOT	0.4454	0.0576	stationary	I(2)*

UNIT ROOT = KPSS rejects H0 (stationarity) at 5% level. p-values bounded at [0.01, 0.10]. Order = final ADF+KPSS consensus classification.

4. Integration Order Summary and VAR/VECM Implications

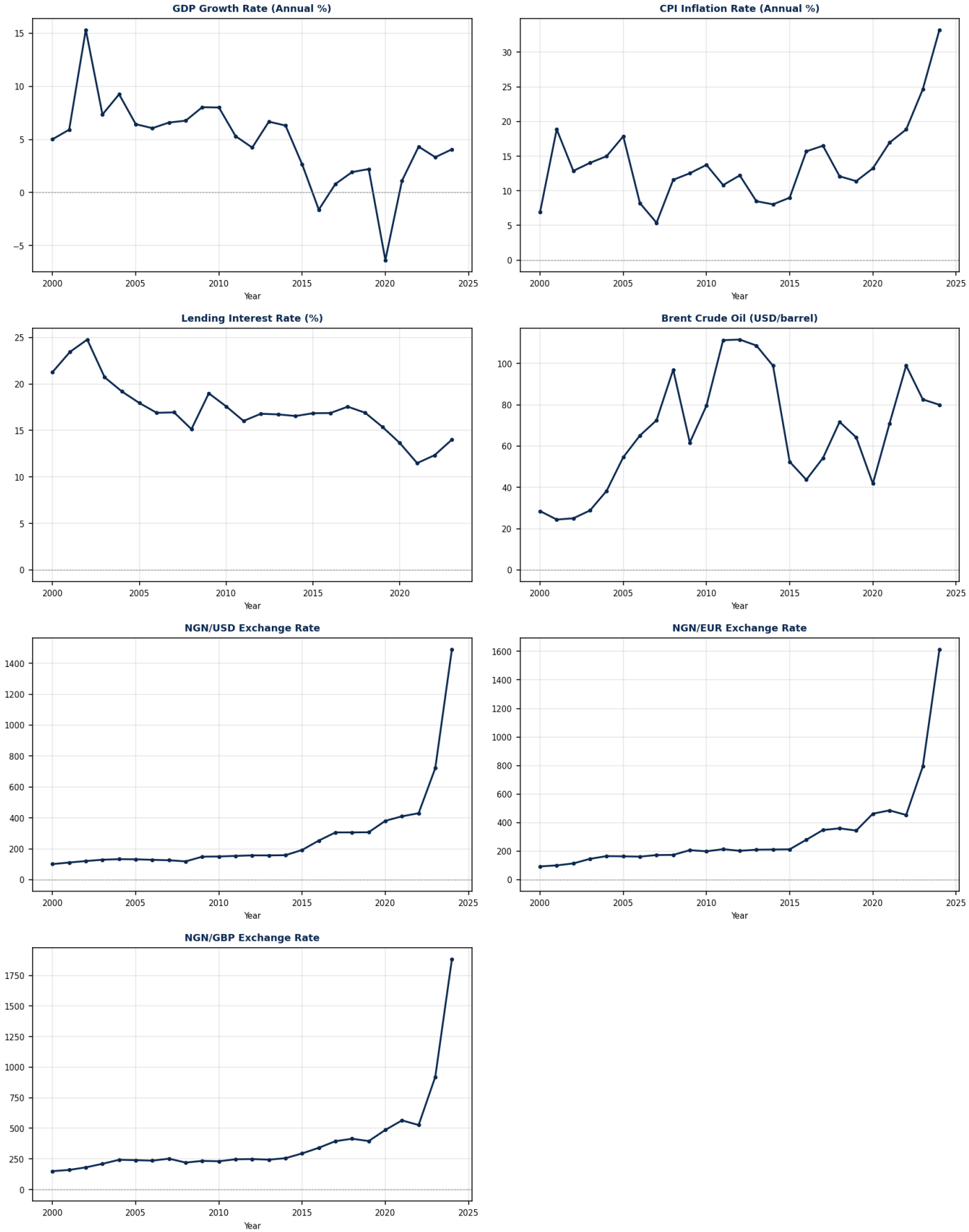
Variable Code	Series Name	Order	VAR/VECM Implication
gdp_growth_pct	GDP Growth Rate (Annual %)	I(1)	Unit root -- 1st-difference in VAR, or include in
cpi_inflation_pct	CPI Inflation Rate (Annual %	I(1)	Unit root -- 1st-difference in VAR, or include in
lending_rate_pct	Lending Interest Rate (%)	I(1)	Unit root -- 1st-difference in VAR, or include in
brent_usd_annual_avg	Brent Crude Oil (USD/barrel)	I(1)	Unit root -- 1st-difference in VAR, or include in
ngn_usd	NGN/USD Exchange Rate	I(2)*	Possibly I(2) -- 2nd-difference or structural brea
ngn_eur	NGN/EUR Exchange Rate	I(2)*	Possibly I(2) -- 2nd-difference or structural brea
ngn_gbp	NGN/GBP Exchange Rate	I(2)*	Possibly I(2) -- 2nd-difference or structural brea

MODEL RECOMMENDATION (O2 Step 3): 4 variables are I(1). Test for Johansen cointegration. If r >= 1 cointegrating vectors found, use VECM to model both short-run dynamics and long-run equilibrium adjustment. The 0 I(0) variable(s) may enter as stationary exogenous regressors.

5. Time Series Charts -- Levels

Trending series (NGN/USD, Brent crude) show clear non-stationarity. GDP growth appears mean-reverting. CPI and lending rates show persistent but potentially trend-stationary behaviour.

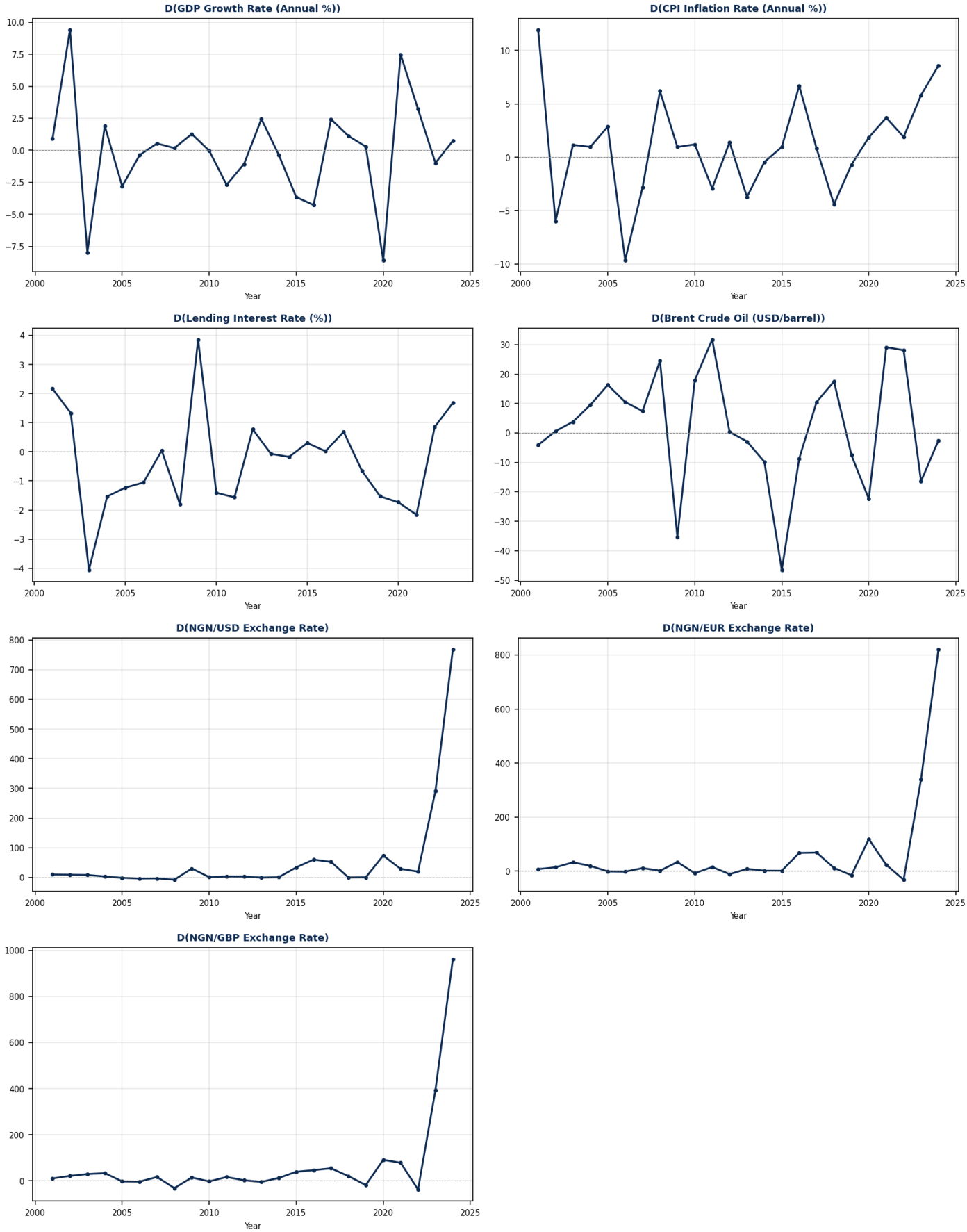
Nigeria Macro Variables -- Levels (2000-2024)



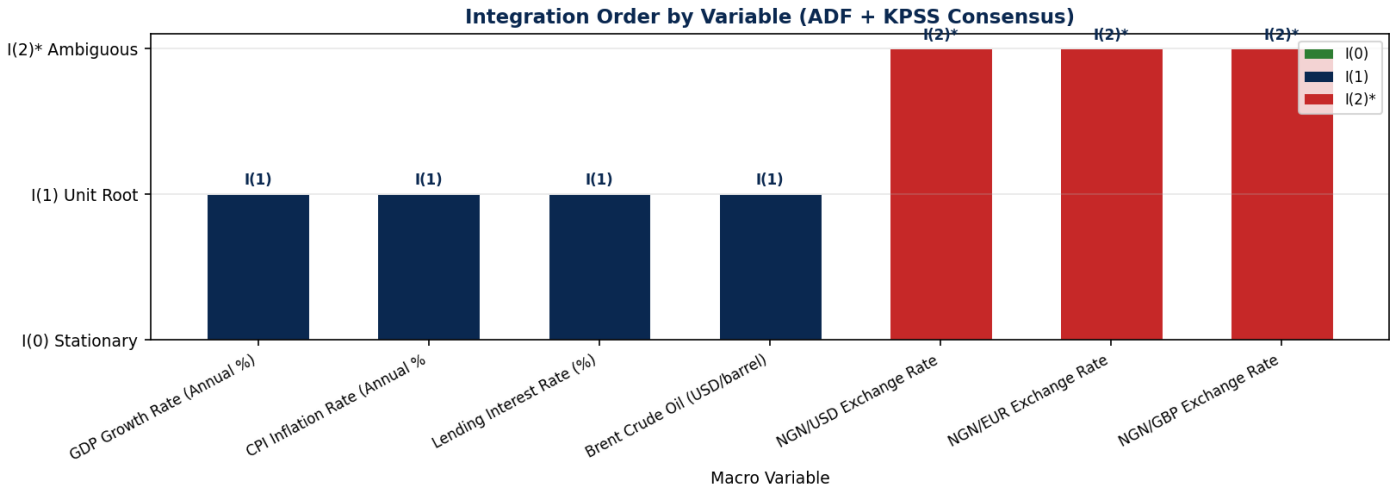
6. Time Series Charts -- First Differences

After differencing, all series fluctuate around zero without persistent trend, supporting I(1) classification for the trending variables.

Nigeria Macro Variables -- First Differences



7. Integration Order Summary Chart



8. Data Quality Notes and Limitations

SMALL SAMPLE: n = 25 annual observations limits test power. ADF power against near-unit-root alternatives is low for n < 50. The dual-test approach and conservative classification partially compensate for this limitation. The final published paper (P3) should apply Ng-Perron (2001) tests and structural break tests (Zivot-Andrews 1992) to address this limitation.

STRUCTURAL BREAKS: Nigeria experienced major economic breaks in this period: 2008 global financial crisis; 2014-2016 oil price collapse; 2016 FX unification; COVID-19 (2020); 2023 petrol subsidy removal and FX liberalisation. Conventional unit root tests have low power in the presence of structural breaks, potentially over-rejecting the stationarity of GDP growth.

9. Next Steps -- O2 Step 3 (VAR/VECM Models)

- Load stationarity_results.csv to determine I(0) vs I(1) split.
- Apply Johansen trace test to I(1) variables (if >= 2 confirmed).
- Select lag order via AIC/BIC (max 2 lags for n=25).
- Fit VECM (if cointegrated) or VAR on differenced I(1) series.
- Produce IRF and FEVD to quantify macro shock transmission.
- Output: O2_05_VAR_VECM_Models.pdf + model_results.csv